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Sub ject : DSA

Unit 2, S10, SLO2 - Doubly Linked List – Insertion at Middle

Algorithm to insert a new node VAL after a given node NUM
Step 1: IF AVAIL = NULL
Write OVERFLOW
Go to Step 12
[END OF IF]
Step 2: SET NEW_NODE = AVAIL
Step 3: SET AVAIL = AVAIL -> NEXT
Step 4: SET ___________________ = VAL
Step 5: SET PTR = START
Step 6: Repeat Step 7 while PTR -> DATA != ________
Step 7: SET PTR = PTR -> NEXT
[END OF LOOP]
Step 8: SET NEW_NODE -> NEXT = _______________
Step 9: SET __________________ = PTR
Step 10: SET _____________ = NEW_NODE
Step 11: SET ______________________ = NEW_NODE
Step 12: EXIT

Answer: Step 4: SET NEW_NODE -> DATA = VAL
Step 6: while PTR -> DATA != NUM
Step 7: SET PTR = PTR -> NEXT
Step 8: SET NEW_NODE -> NEXT = PTR -> NEXT
Step 9: SET NEW_NODE -> PREV = PTR
Step 10: SET PTR -> NEXT -> PREV = NEW_NODE
Step 11: SET PTR -> NEXT = NEW_NODE
Step 12: EXIT

Algorithm to insert a new node VAL before a given node NUM
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[END OF LOOP]
Step 8: SET NEW_NODE -> NEXT = _______
Step 9: SET NEW_NODE -> PREV = ________________
Step 10: SET ______________ = NEW_NODE
Step 11: SET _____________________ = NEW_NODE
Step 12: EXIT

Answer: Step 4: SET NEW_NODE -> DATA = VAL
Step 6: while PTR -> DATA != NUM
Step 7: SET PTR = PTR -> NEXT
Step 8: SET NEW_NODE -> NEXT = PTR
Step 9: SET NEW_NODE -> PREV = PTR -> PREV
Step 10: SET PTR -> PREV -> NEXT = NEW_NODE
Step 11: SET PTR -> PREV = NEW_NODE
Step 12: EXIT